

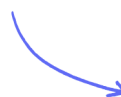
IGCSE • Cambridge (CIE) • Biology

 1 hour  15 questions

Target Test

Created 29/05/26

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Total Marks

/63

- 1** A hare and a rabbit are unable to reproduce to produce fertile offspring.

Which statement explains why?

- A.** they are different species
- B.** they belong to different kingdoms
- C.** they have different habitats
- D.** their DNA base sequences are different

(1 mark)

- 2** The scientific names of some animals are listed.

1. *Camelus dromedarius*

2. *Camelus ferus*

3. *Equus ferus*

4. *Struthio camelus*

Which animals are in the same genus?

- A.** 1, 2 and 3
- B.** 1, 2 and 4
- C.** 1 and 2 only
- D.** 2 and 3 only

(1 mark)

3 (a) Fig. 1 is a photograph of a lion.



Fig. 1

Lions are mammals and have the scientific name *Panthera leo*.

(i)

State **one** feature **visible** in Fig. 1 that identifies the lion as a mammal.

[1]

(ii)

State the genus of this mammal.

[1]

.....
.....
(2 marks)

(b) Mammals are one of the five groups of vertebrates.

Some features of three vertebrate groups are listed. Identify the vertebrate groups.

(i)

Lay soft-shelled eggs

[1]

(ii)

Feathers

[1]

(iii)

Smooth, moist skin

[1]

.....
.....
..... **(3 marks)**

(c) Fig. 2 is a diagram of an animal cell.

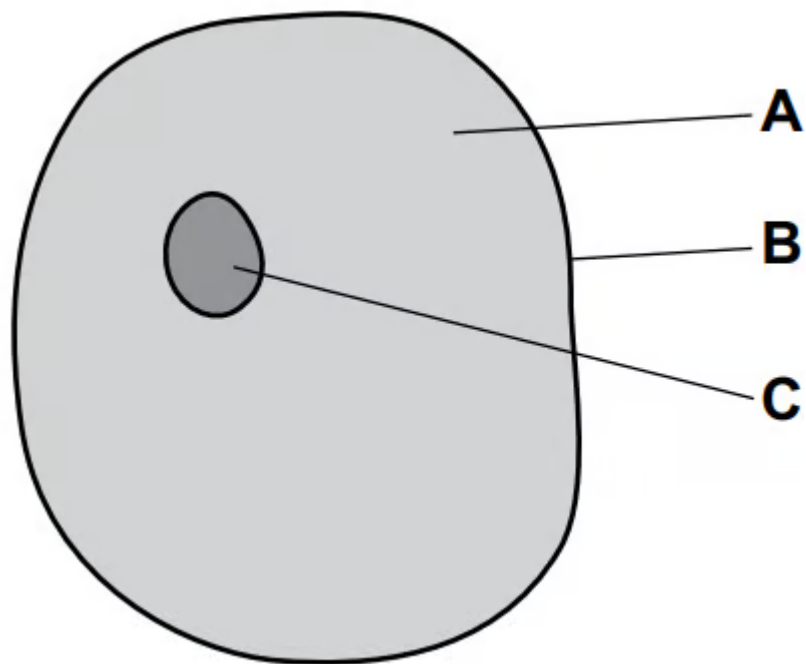


Fig. 2

(i)

Identify the parts labelled **A**, **B** and **C** in Fig. 2

[3]

(ii)

State the names of **two** structures in plant cells that are absent in animal cells.

[2]

(iii)

State the name of **one** structure that is present in bacterial cells and in plant cells but absent in animal cells.

[1]

(6 marks)

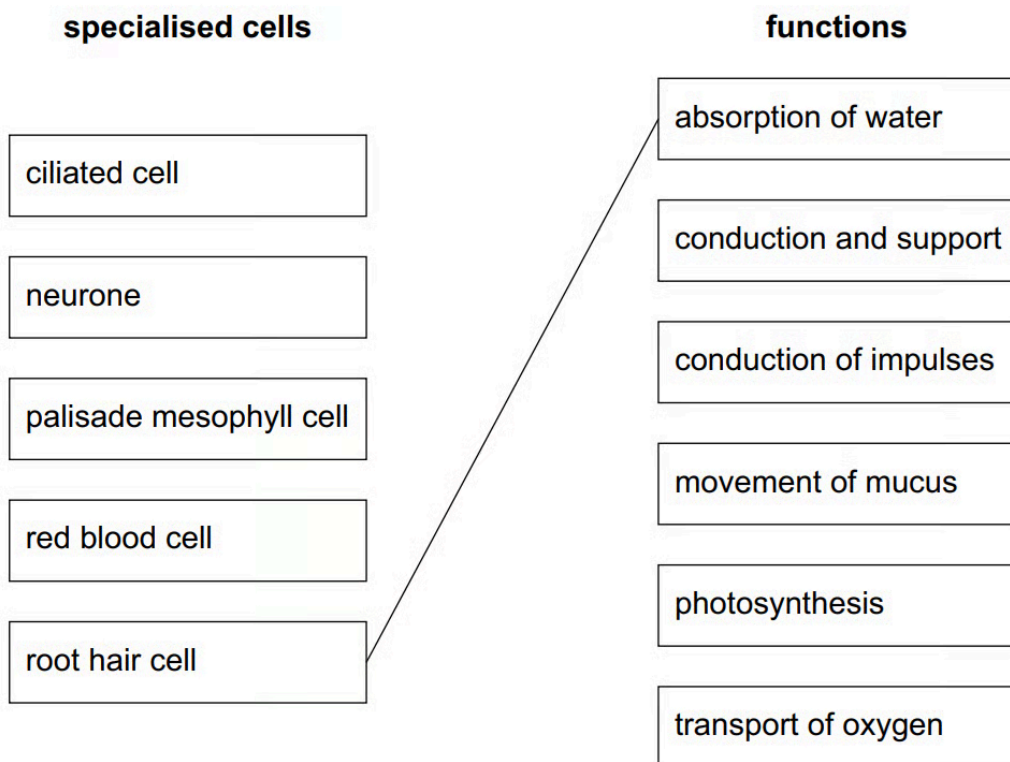
(d) Some cells are specialised to perform a particular function.

The boxes on the left show the names of some specialised cells.

The functions of some specialised cells are in the boxes on the right.

Draw **four** lines to link each specialised cell with its function.

One line has been drawn for you. Draw four additional lines.



(4 marks)

4 (a) A student wanted to investigate the sensitivity of the skin to temperature change.

Fig. 1 shows part of their experimental set up.

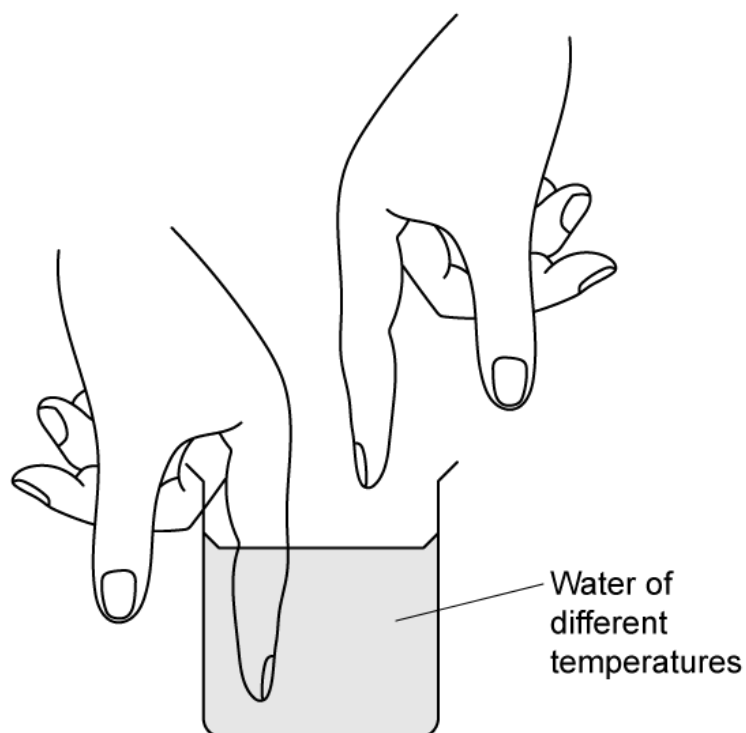


Fig. 1

The students followed the following method:

1. Fill three beakers with water of a different temperatures: one with cold water at 10-15°C, one with warm water at 25°C, and one with hot water at 40-50°C.
2. Place the index finger of the left hand into the cold water and the index finger of the right hand into the hot water. Leave submerged for 2 minutes.
3. After 2 minutes remove both fingers from the hot and cold water and place them repeatedly, but alternately, into and out of the warm for one second at a time. Do this for 1 minute.
4. Record your observations of the sensations felt.

(i)

Identify **one** variable that needs to be kept constant throughout the experiment.

[1]

(ii)

Suggest how the student could make the test more *reliable*.

[1]

(2 marks)

(b) A second student also investigated the sensitivity of the skin. They used a hairpin to determine whether one or two prongs of a hairpin could be felt on different areas of the hand and arm.

1. One student closed their eyes whilst another used the hairpin to gently but firmly touch the skin with either one or two prongs for 1 second.
2. The student with their eyes closed guessed how many prongs were being pressed onto their skin.
3. The test was repeated with increasing gaps of the hair pin prongs on different areas of the hand and arm.

The aim was to determine the most sensitive area of those tested.

Fig. 2 shows the forearm being tested.

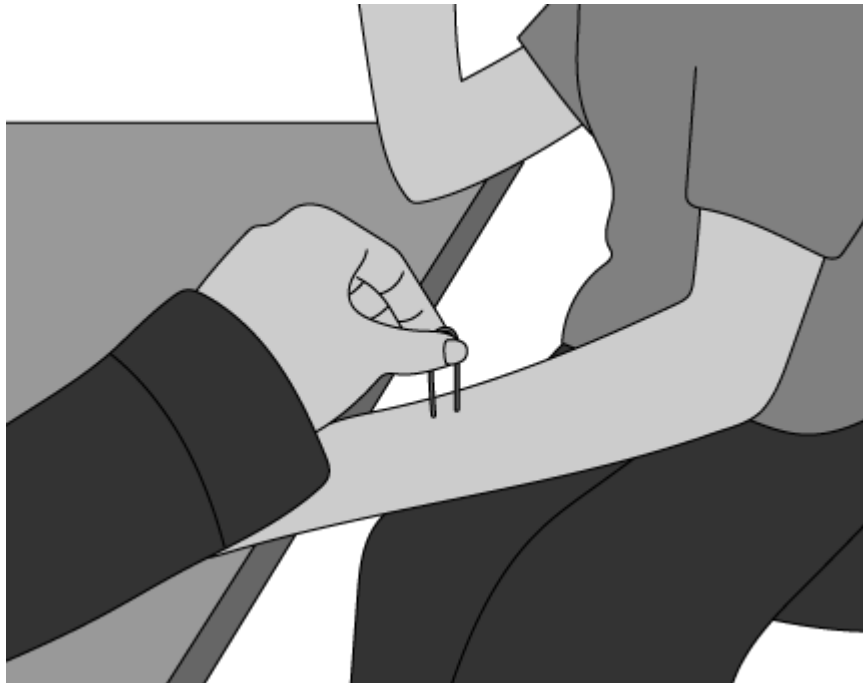


Fig. 2

Their results are shown in Table 1.

Table 1

distance of hairpin prongs / mm	area of arm or hand	number of occasions on which the prong number was detected correctly
5	palm	2
	forearm	1
	elbow	1
10	palm	2
	forearm	2
	elbow	1
15	palm	2
	forearm	2
	elbow	2

(i)

Identify the variable that is measured (the dependent variable).

[1]

(ii)

Describe a source of error associated with the student's experiment.

[1]

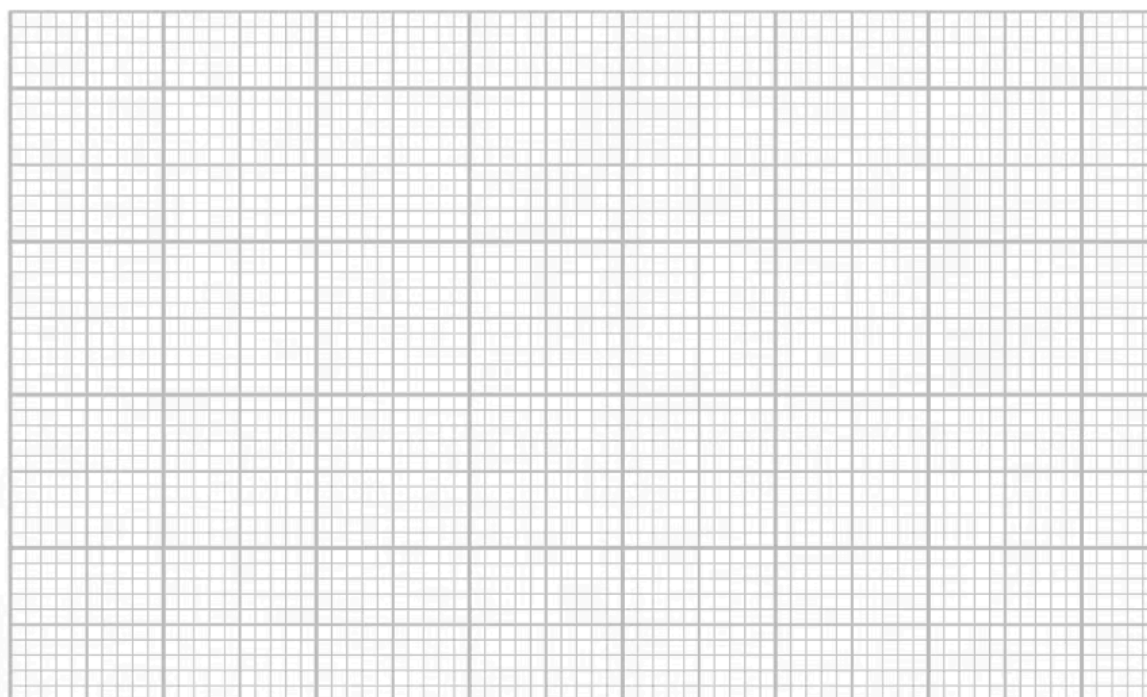
(iii)

Explain improvements that could be made to overcome the source of error.

[1]

(3 marks)

(c) Draw a graph of the students results shown in Table 1.



(4 marks)

5 (a) Extended only

Flowering plants can be divided into two groups.

(i)

State the two groups of flowering plants.

[1]

(ii)

Explain how to distinguish between the two groups.

[3]

(4 marks)

(b) Extended only

A student found the plant shown in Fig. 1 growing in the school grounds.



Fig. 1

Explain what type of plant this would be classified as.

(2 marks)

(c) Extended only

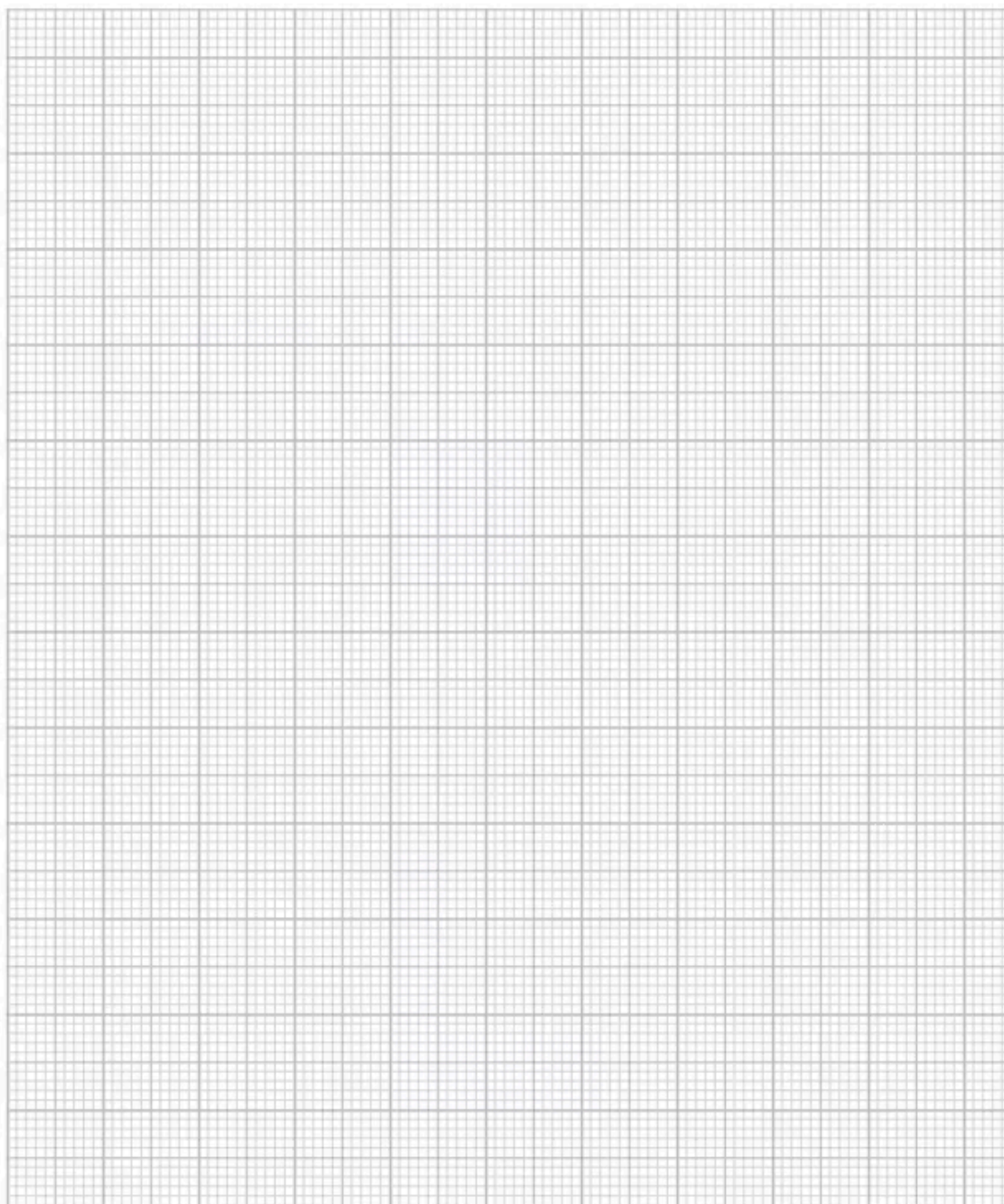
The student investigated how the number of leaves on the plant varied. The student visited three different sites within the school grounds and counted the leaves found on one plant within each site.

Table 1 shows the results of his experiment.

Table 1

school site	number of leaves
A	54
B	23
C	47

Use the grid below to draw a graph of these results.

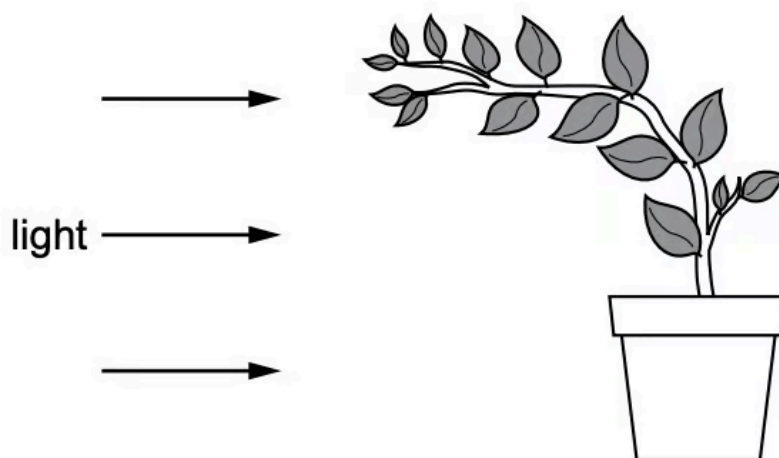


[4]
(4 marks)

(d) **Extended only**

Determine how the student could improve the reliability of their data.

6 The diagram shows a plant.



Which characteristic of living organisms is shown by the plant in the diagram?

- A. Excretion
- B. Reproduction
- C. Respiration
- D. Sensitivity

(1 mark)

7 A scientist observes four events in an organism:

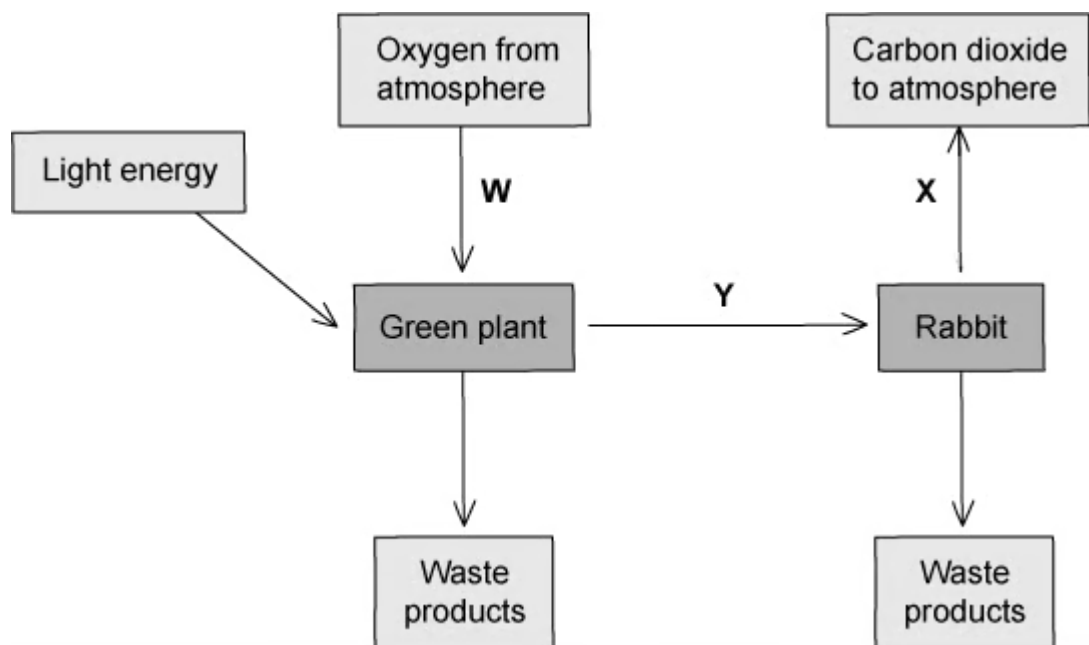
1. It takes in glucose.
2. It releases carbon dioxide produced in cells.
3. It moves away from a harmful chemical.
4. It increases in biomass over time.

Which row correctly matches the events to characteristics?

	1	2	3	4
A	respiration	excretion	sensitivity	growth
B	nutrition	excretion	sensitivity	growth
C	nutrition	respiration	movement	reproduction
D	excretion	nutrition	sensitivity	growth

(1 mark)

8 Some of the processes carried out by living organisms are illustrated in the diagram below.



Which row of the following table correctly describes the characteristics shown by living organisms in the diagram above?

	W	X	Y
A	respiration	photosynthesis	respiration
B	respiration	respiration	nutrition
C	photosynthesis	respiration	excretion
D	photosynthesis	excretion	nutrition

(1 mark)

9 Extended only

The list below shows the binomial names for eight animals.

Canis lupus

Cygnus buccinator

Dasypus novemcinctus

Dasyurus maculatus

Equus burchelli

Eumetopias jubatus

Felis chaus

Felis lybica

Which two animals are most closely related?

A. *Dasypus novemcinctus* & *Dasyurus maculatus*

B. *Felis chaus* & *Felis lybica*

C. *Cygnus buccinator* & *Dasyurus maculatus*

D. *Equus burchelli* & *Eumetopias jubatus*

(1 mark)

10 What are the characteristics of birds?

	skin covered in feathers	maintain a constant body temperature	jelly-covered eggs	external ears present
A	✓	✗	✓	✗
B	✗	✗	✗	✓
C	✓	✓	✗	✗
D	✗	✓	✓	✓

(1 mark)

- 11 The equation below represents a chemical process that occurs in the cells of living organisms.



Which characteristics of living organisms would this equation be associated with?

	nutrition	respiration
A	✓	✓
B	✗	✓
C	✗	✗
D	✓	✗

(1 mark)

12 Lichens are formed from two different organisms living together.

Organism X and organism Y are found in most lichens.

The table shows some of the characteristics of organism X and organism Y.

X	Y
made of strands called hyphae	single-celled
hyphae have cell walls and many nuclei	cell contains a nucleus and chloroplasts

Which kingdoms do X and Y belong to?

	X	Y
A	fungus	prokaryote
B	fungus	protocist
C	protocist	fungus
D	protocist	plant

(1 mark)

- 13** All living organisms release energy from nutrient molecules within their cells.

What is the name of this characteristic?

- A.** Growth
- B.** Nutrition
- C.** Respiration
- D.** Sensitivity

(1 mark)

- 14** The horse, *Equus ferus*, and the donkey, *Equus asinus*, are able to interbreed. The offspring they produce is called a mule.

Which statement is correct?

- A.** The horse and the donkey are the same genus; the mule is infertile.
- B.** The horse and the donkey are the same species; the mule is infertile.
- C.** The horse and the donkey are the same genus; the mule is fertile.
- D.** The horse and the donkey are the same species; the mule is fertile.

(1 mark)

- 15 (a)** Table 1 shows the names of three groups of arthropods and some of their characteristics.

Place ticks (✓) in the boxes to show the characteristics present in each group.

Table 1

characteristic	arthropod group		
	arachnids	crustaceans	myriapods
four pairs of legs			
one pair of antennae			
body divided into two main parts			

(3 marks)

(b) All living organisms show the same seven characteristics.

State **four** of the characteristics of living organisms.

(4 marks)

(c) Fig. 1 shows the apparatus used by some students during an investigation.

The apparatus is drawn as seen from above.

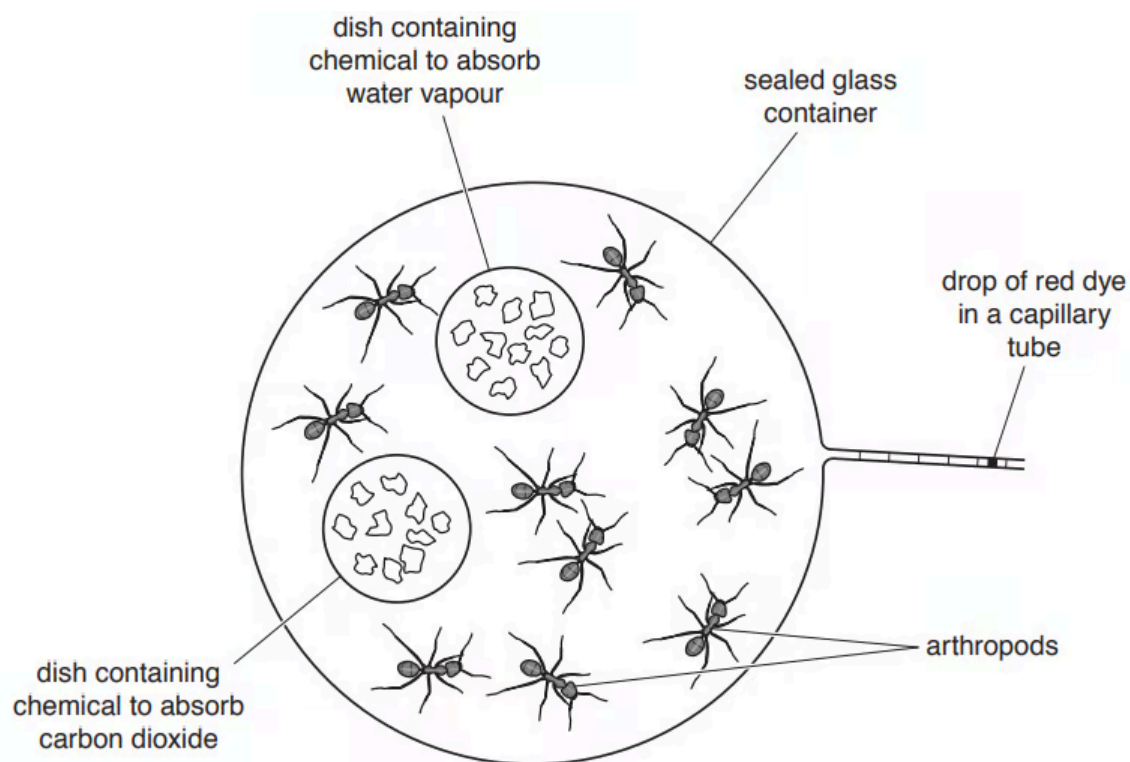


Fig. 1

Some arthropods were placed in the container.

A drop of red dye was inserted into the capillary tube.

The drop of red dye in the capillary tube gradually moved towards the arthropods.

(i)

Explain why the drop of red dye moved towards the arthropods.

The investigation shown in Fig. 1 was repeated at different temperatures.

Fig. 2 shows the results.

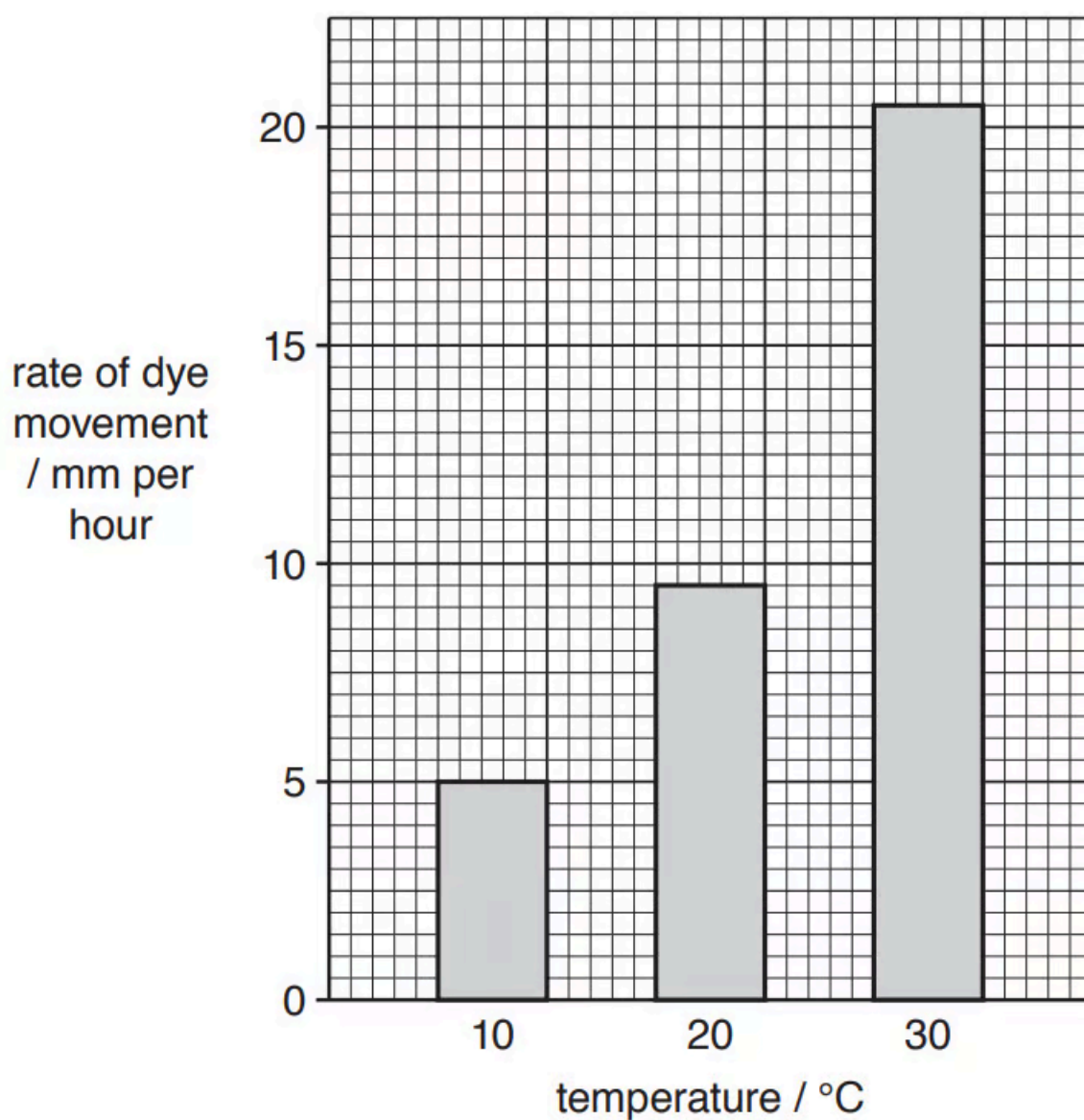


Fig. 2

(ii)

State the trend shown in Fig. 2.

[1]

(iii)

Calculate the percentage increase in the rate of dye movement between 10°C and 20°C.

Show your working.

[2]

(6 marks)

(d) (i)

State the group of arthropods the student used in Fig. 1.

[1]

(ii)

Describe the features visible in Fig. 1 that identify this group.

[3]

(4 marks)